ECA5403-ELECTROMAGNETIC FDIELDS AND WAVES

CO-1

EXPERIMENT 1

AIM

To write a MATLAB program for transformation between Cartesian, Cylindrical and Spherical coordinate systems.

PROGRAM

Test case (i) Transformation of Cartesian coordinate points in to Cylindrical and spherical

coordinate system

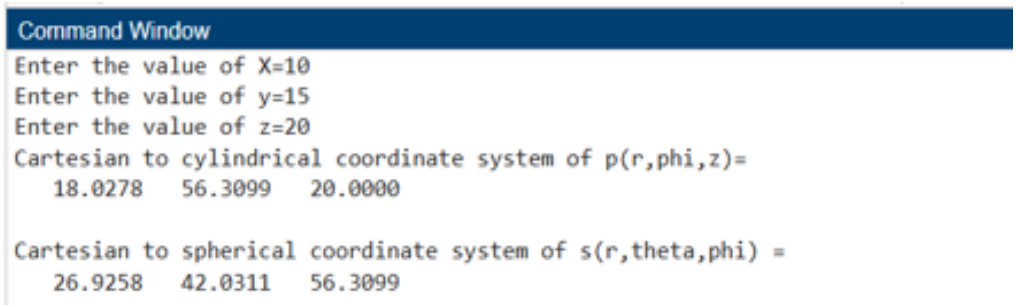
```
clc;
x=input('Enter the value of X=');
y=input('Enter the value of y=');
z=input('Enter the value of z=');

% Cartesian to Cylindrical
r=sqrt(x^2+y^2);
phi=atan2d(y,x);
disp('Cartesian to cylindrical coordinate system of p(r,phi,z)=');
disp([r phi z]);

% Cartesian to Spherical
R=sqrt(x^2+y^2+z^2);
theta=acosd(z/R);
phi=atan2d(y,x);
disp ('Cartesian to spherical coordinate system of s(r,theta,phi) =');
```

disp ([R theta phi]);

OUTPUT



Test case (ii) Transformation of Cylindrical coordinate points into Cartesian and spherical coordinate system

clc;

r=input('Enter the value of r=');

phi=input('Enter the value of phi=');

z=input('Enter the value of z=');

% Cylindrical to Cartesian

x=r*cosd(phi);

y=r*sind(phi);

disp('cylindrical to cartesian coordinate system of p(x,y,z)=');

disp([x y z]);

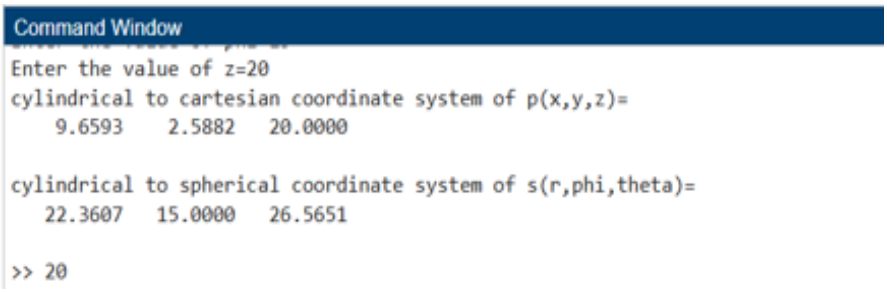
% Cylindrical to Spherical

R=sqrt(r^2+z^2);

theta=acosd(z/R);

disp('cylindrical to spherical coordinate system of s(r,phi,theta)=');

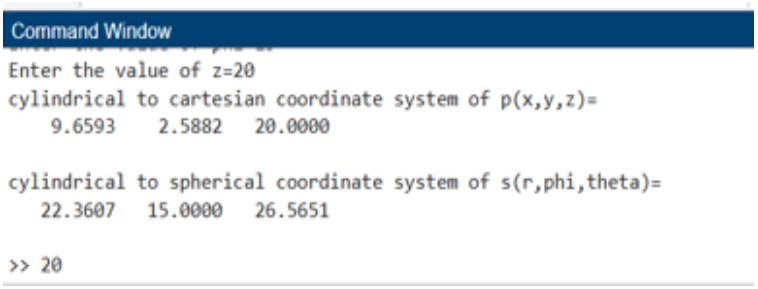
disp([R phi theta]);



Test case (iii) Transformation ofspherical coordinate pointsinto Cylindrical and Cartesian coordinate

system

```
clc;
r=input('Enter the value of r=');
phi=input('Enter the value of phi=');
theta=input('Enter the value of theta=');
% Spherical to Cartesian
x=r*sind(theta)*cosd(phi);
y=r*sind(theta)*sind(phi);
z=r*cosd(theta);
disp('spherical to cartesian coordinate
system of p(x,y,z)=');
disp([x y z]);
% Spherical to Cylindrical
rho=r*sind(theta);
disp('spherical to cylindrical
coordinate system of s(rho,phi,z)=');
disp([rho phi z]);
```



RESULT

Thus the experiment has been done using matlab.

RUBRICS

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Apply coordinate-system definitions and angle conventions	Apply coordinate-system definitions, symbols, units, and angle conventions	Applies coordinate-system definitions and angle conventions throughout.	Uses correct definitions with one minor notation or convention error.	Shows partial understanding; multiple corrections are needed.	Uses incorrect coordinate definitions or angle conventions.	25	2.5	
2	Implement transformations in MATLAB	Implement six transformations in MATLAB	Implements all six transformations correctly using suitable degree-based functions.	Implements most transformations correctly with minor coding errors.	Implements some transformations but needs assistance to correct errors.	Unable to implement the required transformations.	25	2.5	
3	Verify numerical values and reverse transformations	Verify numerical values and reverse transformations	All outputs agree with analytical values; reverse transformations recover	Most values agree; small unexplained differences occur.	Several values differ or reverse verification is incomplete.	Outputs are incorrect or not verified.	30	3	

			the original point within rounding precision.						
4	Interpret and document results	Interpret and document results	Presents complete tables, units, deviations, and a clear conclusion.	Documentation is clear with minor omissions.	Tables or interpretation are incomplete.	Results and conclusion are missing or unclear.	20	2	
Total							100	10	